CPU and decode

File: cpu.kicad_sch

Memory

File: memory.kicad_sch

I/O

File: io.kicad_sch

Power

File: power.kicad_sch

Memory map (24-bit, bank:offset)
  $00:0000-$00:5FFF   RAM  (U5)
  $00:6000-$00:7FFF   I/O window, U6 '138, 8 x 1 KB slots
                      $6000 VIA (U7), $7000 ACIA (U8)
  $00:8000-$00:FFFF   ROM  (U4)
  $01:0000-$07:FFFF   RAM  (U5)
  $08:0000-$7F:FFFF   aliases of banks $00-$07
  $80:0000-$FF:FFFF   VRAMCE decoded; video RAM not fitted (future expansion)

Naming: active-low signals are drawn with an overbar, e.g. $\overline{\text{IOSEL}}$, $\overline{\text{RAMCE}}$, R/$\overline{\text{W}}$.
The W65C datasheets write these with a B suffix (IRQB, RESB, RWB, CS2B);
decode.pld writes them with a leading '/' in its equations.

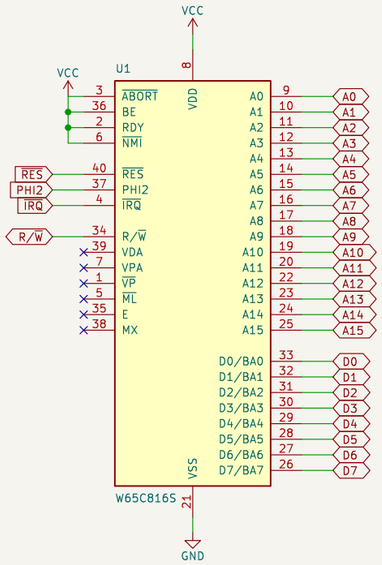
Active-low signals use KiCad overbar notation, e.g. $\overline{\text{IOSEL}}$
W65C816S breadboard computer — see README.md and pld/decode.pld
**be6502**

Sheet: /
File: be6502.kicad_sch

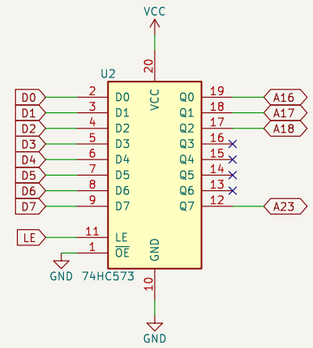
***Title: be6502 — W65C816S breadboard computer***

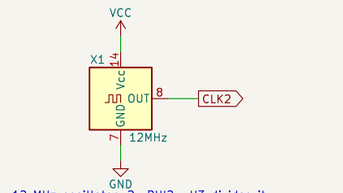
| Size: A4 | Date: 2026-10-16 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 1/5 |

W65C816S (65C816 in DIP-40).
D0-D7 carry the bank byte while PHI2 is low;
U2 captures it while LE is high (see NOTE 6 of decode.pld).
ABORT, BE, RDY and NMI are tied to +5V.
VDA, VPA, VP, ML, E and MX are not connected.

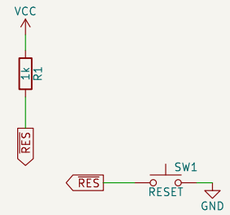


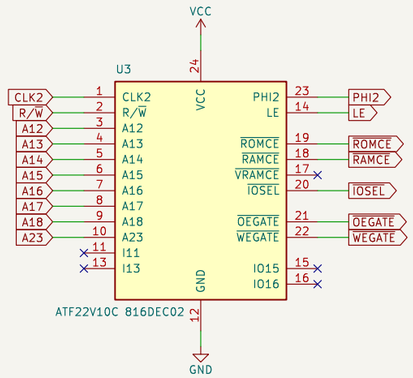

Bank byte latch. Q3-Q6 (A19-A22) are deliberately
not used: banks $08-$7F alias $00-$07 and $81-$FF
alias $80 (decode.pld NOTE 3).




12 MHz oscillator, 2x PHI2. U3 divides it
by two to make PHI2 = 6 MHz. Pin 1 not connected.







IRQ is shared by U7 (VIA) and U8 (ACIA),
both open drain, with pull-up R2.
R1, R2: 1k (estimated; markings unreadable).

U3 runs pld/decode.pld (816DEC02):

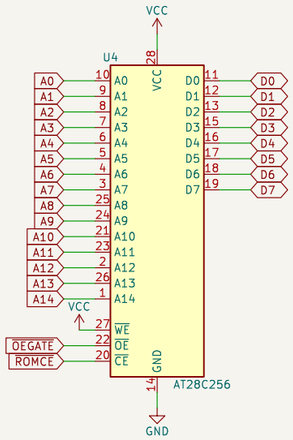
```
  PHI2.R    = /PHI2                (CLK2 / 2)
  LE        = /PHI2 * /CLK2
  /ROMCE    = /A23*/A18*/A17*/A16*A15
  /RAMCE    = bank 0 $0000-$5FFF, or /A23 * (A16+A17+A18)
  /VRAMCE   = A23
  /IOSEL    = /A23*/A18*/A17*/A16*/A15*A14*A13
  /OEGATE   = PHI2 * RWB
  /WEGATE   = PHI2 * /RWB
```

VRAMCE (pin 17) is not connected: video RAM is future expansion.
A12 is wired but unused (NOTE 4). Pins 11, 13, 15, 16 spare.
Burn with power-down disabled (NOTE 5).

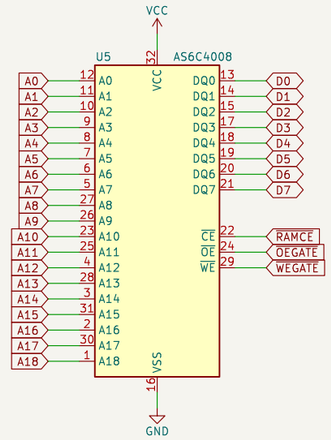
Active-low signals use KiCad overbar notation, e.g. IOSEL
W65C816S breadboard computer - see README.md and pld/decode.pld
**be6502**

| | | |
|---|---|---|
| Sheet: /CPU and decode/ | | |
| File: cpu.kicad_sch | | |
| ***Title: CPU, clock, bank latch and address decode*** | | |
| Size: A3 | Date: 2026-10-16 | **Rev: 1** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 2/5 |

ROM  $00:8000–$00:FFFF  (32 KB)
WE tied high: not writable in circuit.




RAM  512 KB.  A16–A18 come from the bank latch (U2).
$00:0000–$00:5FFF and $01:0000–$07:FFFF reachable.

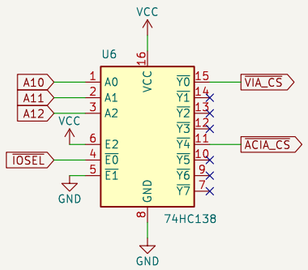


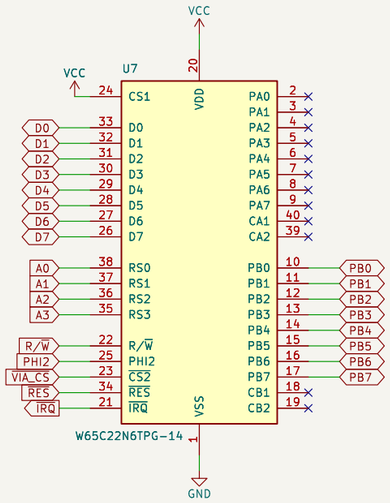

OEGATE is shared by the ROM and RAM; their chip enables are mutually exclusive, so at most one device drives D0–D7 (decode.pld NOTE 1).
WEGATE goes only to the RAM.  It includes PHI2, so no write can happen while D0–D7 carries the bank byte.
Video RAM (62256 on VRAMCE, bank $80) is reserved for future expansion.

Active-low signals use KiCad overbar notation, e.g. IOSEL
W65C816S breadboard computer — see README.md and pld/decode.pld
**be6502**

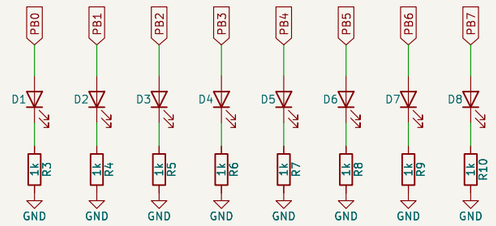
| | | |
|---|---|---|
| Sheet: /Memory/ | | |
| File: memory.kicad_sch | | |
| ***Title: Memory: ROM and RAM*** | | |
| Size: A3 | Date: 2026-10-16 | **Rev: 1** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 3/5 |

I/O window $00:6000–$00:7FFF, 8 x 1 KB slots on A12:A11:A10
  Y0 $6000  VIA        Y4 $7000  ACIA
  Y1 $6400  Y2 $6800  Y3 $6C00
  Y5 $7400  Y6 $7800  Y7 $7C00  (unallocated, not connected)
E1 (pin 5) is the reserved PHI2 hook: tie to PHI2
to qualify the selects later (decode.pld NOTE 2).
E2 (pin 6) is the active-high enable — it must be high.

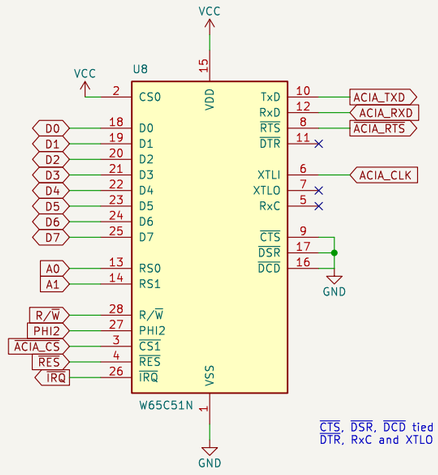


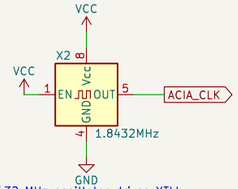

Port A, CA1, CA2, CB1, CB2 not connected.
IRQB is open drain on the W65C22N: wired
straight to IRQ (pull-up R2, CPU sheet).



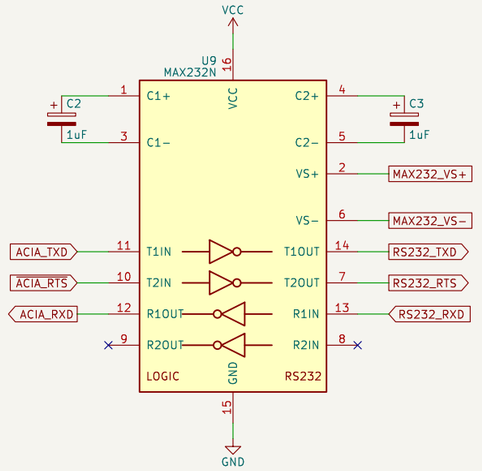

Port B LEDs: PBn -> LED -> resistor -> GND.
R3–R10: 1k (estimated; markings unreadable).

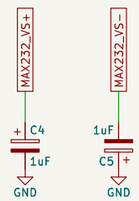


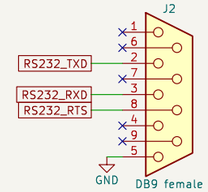

CTS, DSR, DCD tied to GND.
DTR, RxC and XTLO not connected.




1.8432 MHz oscillator drives XTLI.
Pin 1 (EN) tied to +5V.




J2 DB9 female: 2 TxD (T1OUT), 3 RxD (R1IN),
8 RTS from ACIA (T2OUT), 5 GND.
bios.s: 115200 8-N-1, flow control on ACIA RTS.

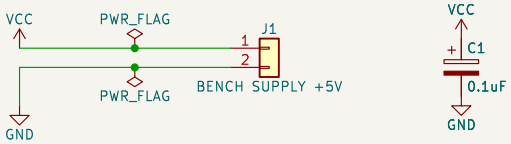
MAX232N: T1 = ACIA TxD, R1 = ACIA RxD, T2 = ACIA RTS.
R2IN and R2OUT not connected.  C2–C5 1uF electrolytic;
C4 from VS+ to GND, C5 from VS− to GND.

Active-low signals use KiCad overbar notation, e.g. IOSEL
W65C816S breadboard computer — see README.md and pld/decode.pld
**be6502**

Sheet: /I/O/
File: io.kicad_sch

**Title: I/O: '138 slot decoder, VIA, ACIA, RS-232**

| Size: A3 | Date: 2026-10-16 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 4/5 |







+5 V from the bench power supply.  C1 (0.1uF electrolytic) between
+5V and GND at the supply connection is the only decoupling capacitor.

Active-low signals use KiCad overbar notation, e.g. $\overline{\text{IOSEL}}$
W65C816S breadboard computer — see README.md and pld/decode.pld
**be6502**

Sheet: /Power/
File: power.kicad_sch

***Title: Power***

| | | |
|---|---|---|
| Size: A4 | Date: 2026-10-16 | **Rev: 1** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 5/5 |